

International Mathematical Olympiad 2024

Hong Kong Team Selection Test 2

21 Oct 2023

Problem 1. Let M be the midpoint of the side BC of an acute $\triangle ABC$, and let D be the foot of perpendicular from C to AM . The circumcircle of $\triangle ABD$ intersects the side BC again at $E \neq B$. Suppose F is a point on the segment AE such that $FB = FC$. Prove that F is the midpoint of AE .

Problem 2. Let n be a positive integer. In each cell of an $n \times n$ grid, we place a coin with either the head or the tail facing upwards. Each time we can select one cell, and flip all the $2n - 1$ coins that lie in the same row or the same column as this cell. Find all n such that it is always possible to have all the coins with the heads facing upwards after finitely many moves, no matter how we place the coins initially.

Problem 3. Let n be a positive integer. Prove that there exists a positive integer m such that

$$3^m + 5^m - 1$$

is divisible by 7^n .

Problem 4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the following condition: for any real numbers x and y , the number $f(x + f(y))$ is equal to $x + f(y)$ or $f(f(x)) + y$.

Time: 3 hours
Each problem is worth 7 points

國際數學奧林匹克 2024
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問題 1. 設 M 是銳角 $\triangle ABC$ 的邊 BC 的中點，並設 D 是 C 到 AM 的垂足。 $\triangle ABD$ 的外接圓與邊 BC 再相交於 $E \neq B$ 。設 F 是線段 AE 上一點，使得 $FB = FC$ 。求證： F 是 AE 的中點。

問題 2. 設 n 是正整數。在一個 $n \times n$ 方格表的每一格中，我們放置一個正面朝上或背面朝上的硬幣。每次我們選擇一個方格，並將這個方格的同一行或同一列的所有 $2n - 1$ 個硬幣反轉。求所有 n ，使得無論我們開始時如何放置硬幣，都總能利用有限次操作令所有硬幣正面朝上。

問題 3. 設 n 是正整數。求證：存在一個正整數 m ，使得

$$3^m + 5^m - 1$$

能被 7^n 整除。

問題 4. 求所有滿足以下條件的函數 $f: \mathbb{R} \rightarrow \mathbb{R}$ ：對任意實數 x 和 y ，都有 $f(x + f(y))$ 等於 $x + f(y)$ 或 $f(f(x)) + y$ 。

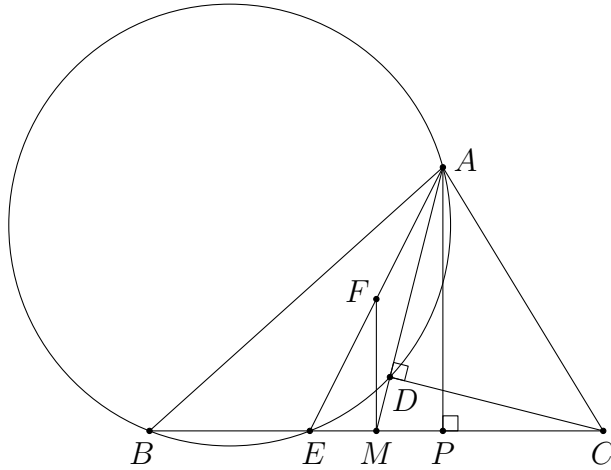
考試時間：3 小時
每題 7 分

Solutions

1. Let P be the foot of perpendicular from A to BC . It follows from $\angle ADC = \angle APC = 90^\circ$ that A, D, P, C are concyclic. Considering the power of M , we find that

$$ME \times MB = MD \times MA = MP \times MC.$$

Since $MB = MC$, we have $ME = MP$. Note that FM is the perpendicular bisector of BC . Therefore, $FM \parallel AP$. As $ME = MP$, it follows from the intercept theorem that F is the midpoint of AE .



2. Answer: 1 and all positive even numbers.

In the following, to operate on the (i, j) th cell (the cell in the i th row and the j th column) means to perform an operation by selecting the (i, j) th cell in the operation.

- When $n = 1$, it is clear that the goal can always be met.
- When n is even, it suffices to show that it is possible to flip any arbitrary coin only in some moves. Indeed, if we want to flip the coin in the (i, j) th cell (called C), we simply operate on every cell that lies in the same row or the same column as C .
 - The coin in C is flipped $2n - 1$ times, which is odd.
 - Each coin in the same row or the same column as C other than the coin in C is flipped n times, which is even.
 - Every other coin is flipped 2 times, which is even.

Therefore, we see that only the coin in C is flipped after these operations.

- When n is odd and greater than 1, it suffices to consider the reverse process, i.e. starting with all coins with the heads facing upwards, we cannot obtain all possible configurations after finitely many moves. Note that there are 2^{n^2} possible configurations since there are two possibilities for each of the n^2 cells. On the other hand, note that the order of operations does not affect the result, and operating the same cell twice does not change the result. Therefore, we may assume each cell is operated 0 or 1 time. This leads to 2^{n^2} different sets of operations. Thus, it suffices to show that there are two different sets of operations that lead to the same configuration.

Indeed, we simply operate on every cell in the first row and the first column except the $(1, 1)$ th cell.

- The coin in the $(1, 1)$ th cell is flipped $2(n - 1)$ times, which is even.
- Each coin in the first row or the first column other than the coin in the $(1, 1)$ th cell is flipped $n - 1$ times, which is even.
- Every other coin is flipped 2 times, which is even.

Therefore, we see that no coin is flipped after these operations, which means this has the same effect as not performing any operation at all (which is also one of the 2^{n^2} sets of operations). This completes the proof.

3. We show that $m = 7^{n-1}$ works. By the lifting the exponent lemma, we have

$$v_7(3^m + 4^m) = v_7(3 + 4) + v_7(m) = 1 + (n - 1) = n.$$

This implies $7^n \mid 3^m + 4^m$, and hence $3^m \equiv -4^m \pmod{7^n}$. Similarly, $5^m \equiv -2^m \pmod{7^n}$. It follows that

$$3^m + 5^m - 1 \equiv -4^m - 2^m - 1 = -(2^{2m} + 2^m + 1) \pmod{7^n}. \quad (1)$$

Next, again by the lifting the exponent lemma, we have

$$v_7(8^m - 1) = v_7(8 - 1) + v_7(m) = 1 + (n - 1) = n.$$

This implies $7^n \mid 8^m - 1 = (2^m - 1)(2^{2m} + 2^m + 1)$. As $3 \nmid m$, we have $7 \nmid 2^m - 1$. Therefore, we must have

$$7^n \mid 2^{2m} + 2^m + 1. \quad (2)$$

Combining (1) and (2), we see that $7^n \mid 3^m + 5^m - 1$ as desired.

Remarks. The use of the lifting the exponent lemma is not necessary. In fact, we only need to prove 7^n divides each of $3^m + 4^m$, $5^m + 2^m$ and $8^m - 1$ in the above proof, all of which can be proved by induction on n .

4. Answer: $f(x) = x + c$ for some constant c .

We first check that all such functions are solutions. Indeed, if $f(x) = x + c$, then

$$f(x + f(y)) = f(x + y + c) = x + y + 2c = f(f(x)) + y$$

for any $x, y \in \mathbb{R}$.

Next, we prove that these are the only solutions. Replacing x by $x - f(y)$, the given condition means

$$f(x) = x \quad \text{or} \quad f(x) = f(f(x - f(y))) + y$$

for any $x, y \in \mathbb{R}$. We are done if $f(x) = x$ for all $x \in \mathbb{R}$. Assume $f(a) \neq a$ for some $a \in \mathbb{R}$. Then we have

$$f(a) = f(f(a - f(y))) + y \tag{1}$$

for any $y \in \mathbb{R}$. Note that f is injective, since if $f(\alpha) = f(\beta)$, then

$$\alpha = f(a) - f(f(a - f(\alpha))) = f(a) - f(f(a - f(\beta))) = \beta$$

by (1). Setting $y = 0$ in (1), we have $f(a) = f(f(a - f(0)))$, and hence

$$a = f(a - f(0)) \tag{2}$$

as f is injective. In particular, this implies $f(0) \neq 0$ (as $f(a) \neq a$).

- If $f(x) \neq x$ for all $x \in \mathbb{R}$, then (2) holds for all $a \in \mathbb{R}$. Replacing a by $x + f(0)$, this gives $f(x) = x + c$ for all $x \in \mathbb{R}$, where $c = f(0)$ is a constant as desired.
- Suppose $f(b) = b$ for some b . Note that $f(f(a)) \neq f(a)$ since $f(a) \neq a$ and f is injective. It follows that (1) holds if we replace a by $f(a)$. Thus, we obtain

$$f(f(a)) = f(f(f(a) - f(y))) + y.$$

Putting $y = a$, this yields

$$f(f(a)) = f(f(0)) + a. \tag{3}$$

Next, putting $y = b$ in (1), we have $f(a) = f(f(a - b)) + b$. Recall that $f(a) \neq a$, so we cannot have $f(a - b) = a - b$. Therefore, using (3) with a replaced by $a - b$, we deduce

$$f(a) = f(f(a - b)) + b = f(f(0)) + (a - b) + b = a + f(f(0)). \tag{4}$$

Recall that $f(0) \neq 0$. Thus, we may take $a = 0$ in (4). This implies $f(0) = f(f(0))$. But then this yields $0 = f(0)$ by the injectivity, which is a contradiction.